

TAO (TONY) JIN

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PROFESSIONAL SUMMARY

Data Analyst with a Cornell M.Eng. in Systems Engineering (May 2026) and a UBC B.Sc. in Mathematics, plus three data, analytics, and financial research internships across North America and Asia. Builds production SQL, automated Extract-Transform-Load (ETL) workflows, business intelligence (BI) dashboards, and KPI reporting that stakeholders use to make decisions. Strengths in SQL, Python, R, Excel, statistical analysis, A/B testing, cohort and retention analysis, and data storytelling, with a record of translating ambiguous business questions into analytical solutions and presenting findings to non-technical stakeholders. Fluent in English and Mandarin Chinese.

TECHNICAL SKILLS

Querying & Databases: SQL (PostgreSQL, Oracle SQL, DuckDB) — joins, CTEs, window functions, aggregations, subqueries, query optimization; relational database design, data modeling, database normalization (3NF), data marts, data warehousing concepts

Business Intelligence & Visualization: Tableau, Metabase, Excel (pivot tables, lookup formulas, dashboards, scorecards), Matplotlib, dashboard development, KPI reporting, executive reporting, data visualization, data storytelling

Programming & Analysis: Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib), R, C++, MATLAB, LaTeX, Jupyter, Git/GitHub version control

Statistics & Analytics Methods: Exploratory Data Analysis (EDA), descriptive and inferential statistics, hypothesis testing, statistical significance, A/B testing and experimental design, regression analysis, statistical modeling, time series forecasting (SARIMA, Holt-Winters), cohort analysis, customer segmentation, retention and churn analysis, root cause analysis, correlation analysis, optimization and linear programming

Data Engineering & Quality: ETL/ELT pipeline development (Mage), workflow automation, data cleaning and wrangling, data validation, data quality auditing, data integrity, reproducible analysis, technical documentation, REST API integration, JSON

Machine Learning & AI: Supervised learning, classification, predictive modeling, feature engineering, model evaluation (AUROC, AUPRC, precision, recall), cross-validation, model calibration, conformal prediction, SHAP explainability, subgroup fairness and bias auditing, Natural Language Processing (NLP), Large Language Models (LLMs), LLM APIs, prompt engineering, structured output extraction

Business & Communication: Stakeholder communication and management, cross-functional collaboration, requirements gathering, business acumen, presenting to technical and non-technical audiences, technical writing, data-driven decision making, attention to detail, working through ambiguity

Languages: Mandarin Chinese (Native), English (Professional Working Proficiency), Japanese (Basic)

EDUCATION

Cornell University

Master of Engineering in Systems Engineering

Relevant Coursework: Data Analytics, Optimization, Statistical Modeling, Machine Learning

Ithaca, NY

Expected May 2026

University of British Columbia

Bachelor of Science, Major in Mathematics

Relevant Coursework: Statistical Learning, Time Series Analysis, Statistical Inference & Hypothesis Testing, Stochastic Calculus, Linear Programming, Algorithms & Data Structures, Relational Databases

Vancouver, BC

Sep. 2021 – June 2025

PROFESSIONAL EXPERIENCE

Aifuku

Data Science Intern

Toronto, ON

Jan. 2025 – May 2025

- Partnered cross-functionally with the operations team to write and optimize PostgreSQL queries (joins, CTEs, aggregations) against a Postgres 15 source-of-truth database of 9 core tables across 4 operational inputs, validating order, receipt, inventory, and ledger data feeding Metabase business intelligence (BI) dashboards and KPI reporting.
- Built and documented a 2-block Mage ETL pipeline that retrieved the latest bank and credit-card statement PDFs from Google Drive, applied a Gemini Large Language Model (LLM) API with prompt engineering to parse transaction fields, and output structured JSON for ledger reconciliation — replacing manual data entry with a reproducible, automated workflow.
- Performed exploratory data analysis (EDA) across 5 linked operational entities — receipts, line items, inventory movements, production batches, and customer orders — running root cause analysis on data quality and schema issues, then presented findings and recommendations to non-technical operations stakeholders to improve reporting accuracy and data integrity.

Grow Investment Group

Institutional Sales Department Intern

Shanghai, China
May 2024 – Aug. 2024

- Engineered Python (pandas) automation scripts to cross-reference multi-source databases during institutional client due diligence and due diligence questionnaire (DDQ) completion, reducing manual data processing time by 15% while improving data accuracy through systematic validation checks.
- Conducted quantitative market research and statistical analysis integrating macroeconomic indicators, interest rate trends, and market sentiment data; collaborated with senior analysts to author written analytical reports supporting institutional investment decisions.
- Authored a convertible bond market downturn analysis and product training materials for the client-facing sales team, translating quantitative findings into clear data storytelling for a non-technical audience; contributed to a 9% reduction in client redemptions, 4% more client inquiries, and a 12% increase in conversion rates.

Sinolink Securities Co., Ltd.

Investment Banking Intern

Shanghai, China
June 2023 – Aug. 2023

- Calculated and visualized portfolio performance and risk metrics (Sharpe ratio, Sortino ratio, Beta, volatility) and ran correlation analysis, benchmarking trends against market indices in risk management reports used by internal decision-makers.
- Performed pre-IPO financial due diligence for a Shanghai main-board applicant (planned fundraising ~RMB 460M), reconciling a dozen executives' 3-year bank statements against company ledgers with rigorous attention to detail to flag unexplained transactions and disclosure discrepancies.
- Redesigned data collection, spreadsheet verification, and reporting workflows through cross-functional collaboration with senior analysts, eliminating duplicate data entry and improving team operational efficiency by 8%; standardized and documented the workflow for team reuse.

PROJECT EXPERIENCE

E-Commerce Marketplace Retention & Fulfillment Analytics (Olist)

SQL (DuckDB), CTEs, Window Functions, Cohort Analysis, Excel Dashboard

- Designed reusable order-, customer-, and item-level data marts in DuckDB SQL from 99,441 marketplace orders, aggregating payments, items, and reviews to order grain before joining — a data modeling decision that prevented double-counted revenue and protected KPI accuracy.
- Ran cohort analysis, customer segmentation, and retention analysis to surface a 3.0% delivered-customer repeat-purchase rate and a BRL 7.98M high-value one-time-buyer segment, converting the insight into CRM and second-purchase campaign recommendations.
- Analyzed 96,478 delivered orders to quantify an 8.1% late-delivery rate and its materially weaker review scores, then built an executive-ready recommendation scorecard and Excel dashboard covering service recovery, seller risk, and category operations.

Diabetes 30-Day Hospital Readmission Risk Model

Sep. 2025 – Dec. 2025

Python, pandas, scikit-learn, XGBoost, SHAP

- Engineered a 160-feature clinical dataset from 101,766 hospital encounters covering 71,518 unique patients across 130 U.S. hospitals — spanning diagnoses, medications, and utilization metrics — applying patient-level 70/10/20 train-calibration-test splits to prevent data leakage in a reproducible modeling pipeline.
- Benchmarked logistic regression, random forest, gradient boosting, and XGBoost classifiers with cross-validation for imbalanced binary classification (best held-out AUROC 0.668, AUPRC 0.230), then applied isotonic recalibration to cut Expected Calibration Error from 0.342 to 0.004 and ran SHAP interpretability plus subgroup fairness auditing.

LLM Metadata Extraction Pipeline — Kaggle Competition

Jan. 2026 – Mar. 2026

Python, NLP, LLM APIs, pandas, REST API

- Built an automated Natural Language Processing (NLP) pipeline extracting structured SDRF metadata across a 71-category annotation taxonomy from 107 scientific publications, processing 1,659 sample records over 15 test datasets.
- Reverse-engineered the competition scoring function (difflib string-similarity clustering at a 0.80 threshold) to root-cause a 0.575 validation-to-test performance gap, isolating vocabulary overfitting and value normalization — not extraction logic — as the primary score driver.

Additional Projects: Time Series Forecasting — Canadian Labor Hours (R, SARIMA, Holt-Winters, AIC parameter tuning, 95% prediction intervals for stakeholders) · Dual-Portal Delivery Platform Database Design (Oracle SQL, Node.js, 3NF normalization, referential integrity, real-time synchronization).